



T503 Wi-Fi Module V1.0.0

1. Product Overview

T503 is a low-power embedded Wi-Fi module. It consists of a highly integrated wireless radio frequency chip LN8825B and a small number of peripheral devices. It has a built-in Wi-Fi network protocol stack and rich library functions. T503 has built-in ARM-CM4 MCU, 2Mbyte flash memory, 296Kbyte SRAM and rich peripheral resources.

T503 runs RTOS platform and integrates all Wi-Fi MAC and TCP/IP protocol function libraries. Users can develop their own Embedded Wi-Fi products that meet your needs.

The LN8825B block diagram is shown in Figure 1:

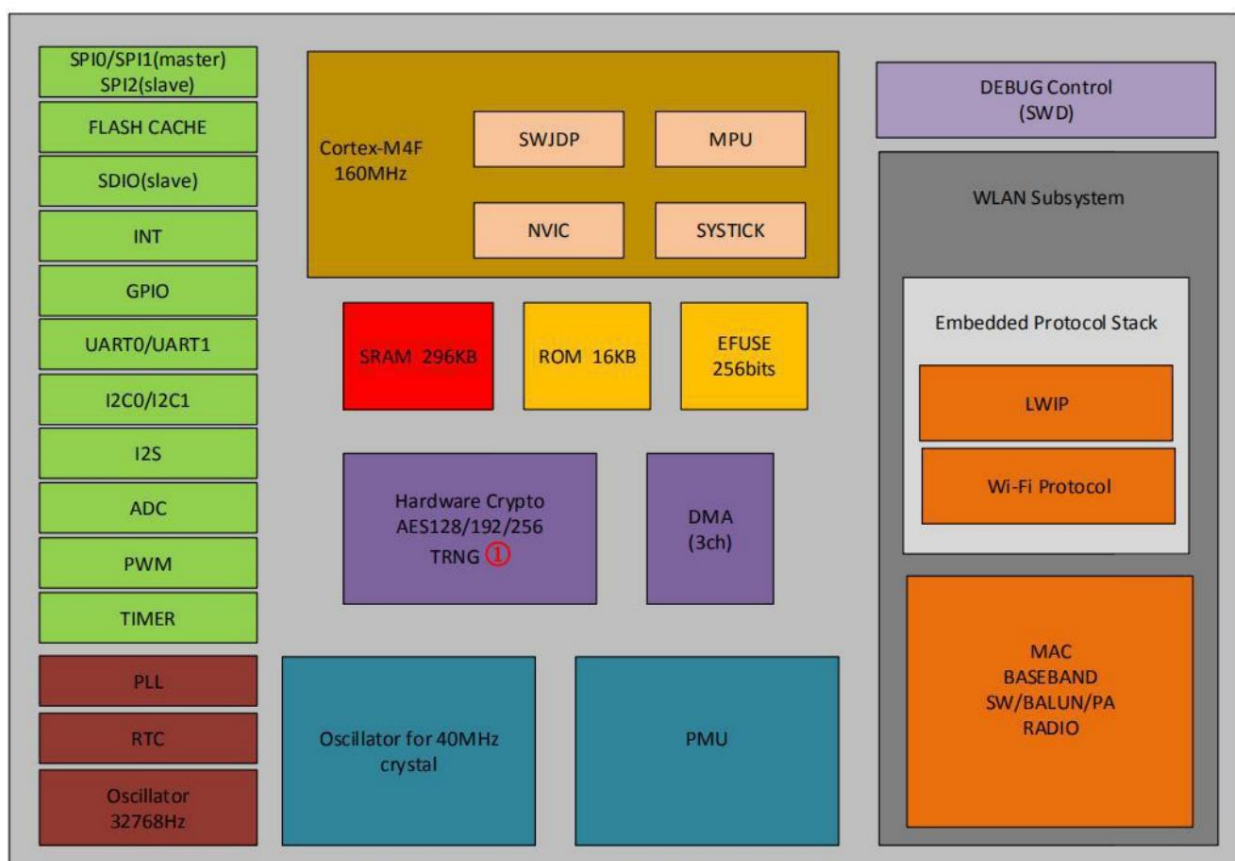


Figure 1 Block diagram

1.1 Features

• Built-in ARM_CM4 MCU, can double as application processor •

Working voltage: 3.3V • Wi-

Fi connectivity • 802.11

b/g/n • Channel

1-14@2.4GHz • Support WPA/

WPA2 security mode • Maximum in 802.11b

mode 18dBm output power • Support STA/AP/STA+AP working

mode

• Support SmartConfig function (including Android and IOS devices) • Onboard
PCB antenna • Operating
temperature: -20° ~ 85°

1.2 Main application areas

• Smart buildings
• Smart home/home
appliances • Smart sockets,
smart lights • Industrial
wireless control •
Baby monitors •
Network cameras • Smart buses

1.3 Physical picture

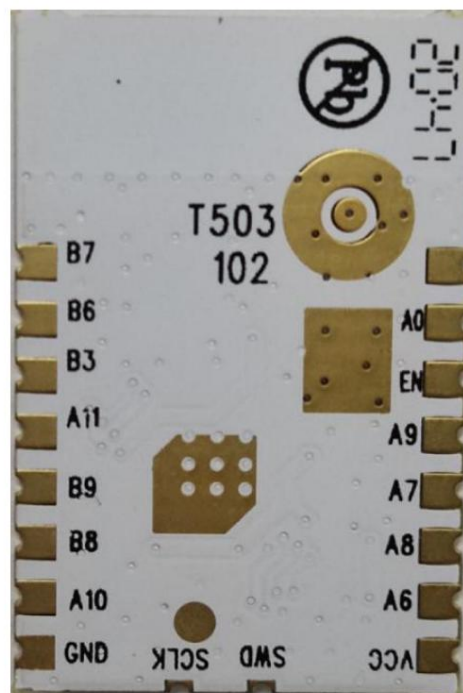


Figure 2 T503| Rear view

2. Module interface

2.1 Pin definition

T503 has 2 rows of pins with a pin spacing of 2mm.

T503 dimensions: 16mm (W) × 24mm (L) × 0.8mm (H). T503 pin definition is shown in Figure 2:

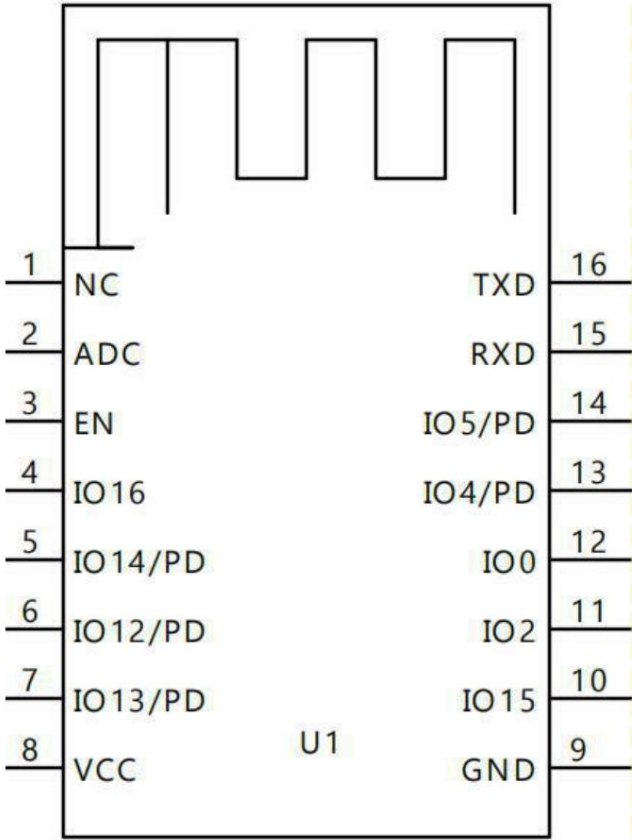


Figure 3 T503 pin diagram

The interface pin definition is shown in Table 1:

Table 1 T503 interface pin arrangement description

serial number	symbol	IO Type	Function
1	NC	/	NC
2	ADC	AI	ADC input pin (3.3V) (1)
3	IN	I	chip enable pin (internal pull-up resistor, low level shutdown)
4	A9	I/O	GPIOA_A9/SDIO_CLK
5	A7	I/O	GPIO_A7/SDIO_D3
6	A8	I/O	GPIO_A8/SDIO_CMD
7	A6	I/O	GPIO_A6/SDIO_D2

8	VCC	P module power pin (3.3V)
9	GND	P Power reference ground
10	A10	I/O GPIO_A10/SDIO_D0
11	B8	I/O GPIO_B8/U0RXD
12	B9	I/O GPIO_B9/U0TXD
13	A11	I/O GPIO_A11
14	B3	I/O GPIO_B3
15	B6	I/O GPIO_B6/U1RX
16	B7	I/O GPIO_B7/U1TX

Note: P indicates the power pin, I/O indicates the input/output pin, and AI indicates the analog input pin.

EN is only the module hardware enable pin and cannot clear the Wi-Fi network configuration information.

(1): This pin can only be used as an ADC port and cannot be used as an ordinary IO port. If this port is not used, it must be left floating.

3. Electrical parameters

3.1 Absolute electrical parameters

Table 2 Absolute parameters

parameter		Minimum	maximum value	unit
Ts	Description	-20	85	°C
VCC	Storage Temperature Supply Voltage	-0.3	3.6	V

3.2 Working conditions

Table 3 Normal working conditions

parameter	Description	Minimum	Typical Value	maximum value	unit
T _{oper}	Operating Temperature	-20	-	85	°C
VCC	Operating Voltage	3.0	3.3	3.6	V
V _{IL}	IO low level input	-0.3	-	VCC*0.25	V
V _{IH}	IO high level input	VCC*0.75	-	VCC	V
V _{OL}	IO low level output	-	-	VCC*0.1	V
V _{OH}	IO high level output	VCC*0.8	-	VCC	V
I _{max}	IO drive current	-	-	16	mA

3.3 Wi-Fi Transmit Power Consumption

Table 4 TX power consumption when continuously transmitting

symbol	parameter			Typical value	unit
	model	rate	Transmit power		

IRF	11b	11Mbps	18dBm	260	mA
IRF	11g	54Mbps	14dBm	220	mA
IRF	11n	MCS7	13dBm	220	mA

3.4 Wi-Fi reception power consumption

Table 5 RX power consumption during continuous reception

symbol	parameter		Typical Value	unit
	model	rate		
IRF	11b	11Mbps	63	mA
IRF	11g	54Mbps	68	mA
IRF	11n	MCS7	68	mA

4. RF characteristics**4.1 Basic RF characteristics**

Table 6 Basic RF characteristics

Parameters	Detailed description
working frequency	2.412~2.484GHz
Wi-Fi Standards	IEEE 802.11b/g/n (channel 1-14)
data transfer rate	11b:1,2,5.5,11 (Mbps) 11g:6,9,12,18,24,36,48,54(Mbps) 11n:BW20_MCS7 65Mbps
Antenna Type	PCB antenna (default), gain 1dBm

4.2 Wi-Fi Output Power

Table 7 TX continuous transmission power

parameter		Minimum value	Typical value	Maximum value	unit
model	rate				dBm
RF average output power, 802.11b CCK Mode	11M	-	18	-	dBm
RF Average Output Power, 802.11g OFDM Mode	54M	-	14	-	dBm
RF average output power, 802.11n OFDM Mode	BW20_MCS7	-	13	-	dBm
frequency error		-10	-	10	ppm

4.3 Wi-Fi reception sensitivity

Table 8RX sensitivity

parameter		Min.	Typ.	Max.	unit
model	rate				dBm
PER<8%, RX sensitivity, 802.11b CCK Mode	11M	-	-88	-	dBm
PER<10% RX sensitivity, 802.11g OFDM Mode	54M	-	-74	-	dBm

PER<10% RX sensitivity, 802.11n OFDM Mode	BW20_MCS7		-71		dBm
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5. Antenna information

5.1 Antenna types

Only PCB on-board antenna access method.

5.2 Reduce antenna interference

When using the PCB onboard antenna on the Wi-Fi module, to ensure optimal Wi-Fi performance, it is recommended to

The distance between the parts and other metal parts should be at least 15mm.

6. Packaging information and production instructions

6.1 Mechanical dimensions

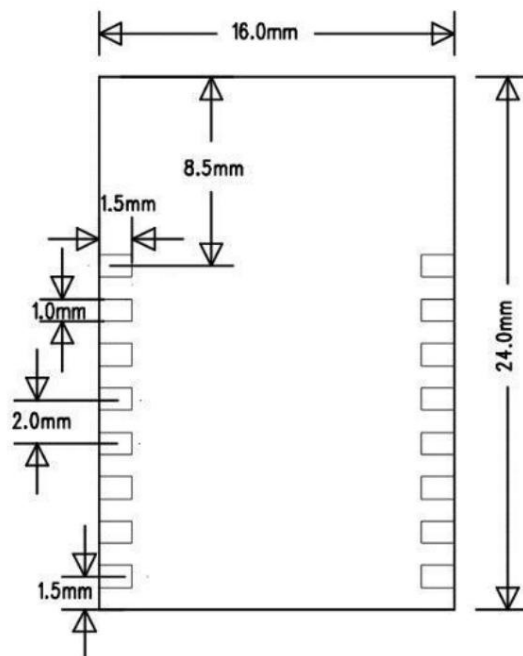


Figure 4 T503 dimensions

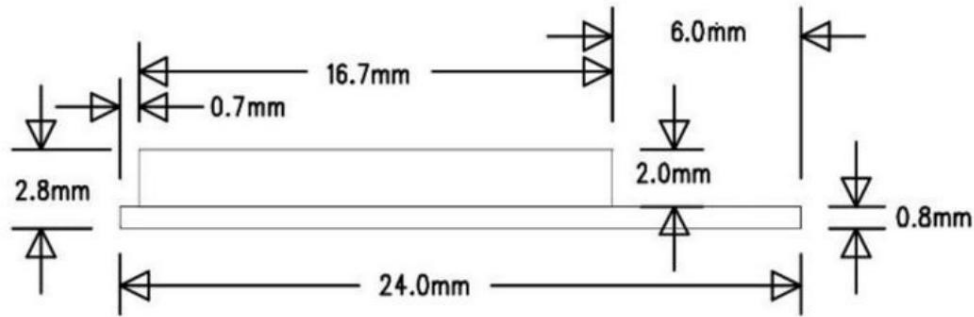


Figure 5 T503 side view

6.2 PCB Recommended Package

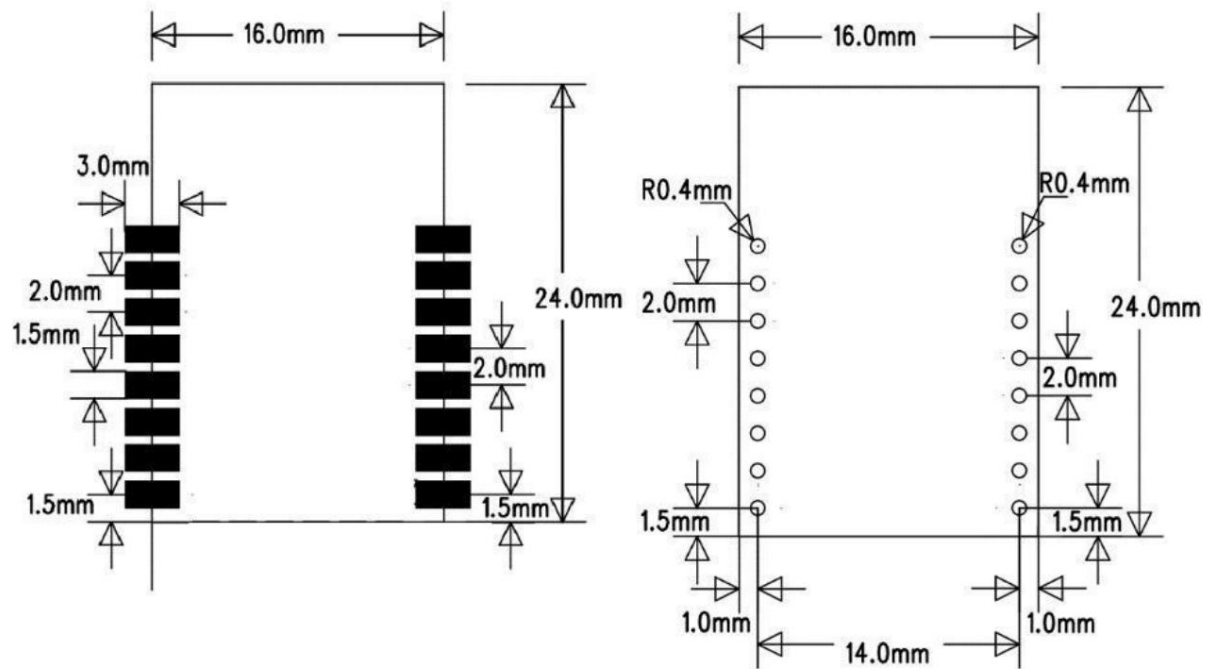


Figure 6 T503 PCB SMD package size | Plug-in package size

6.3 Production Guidelines

The storage conditions of the modules shipped

from the factory are as follows: 1. Moisture-proof bags must be stored in an environment with a

temperature <30°C and a humidity <85%RH. 2. The shelf life of dry-packaged products should be 6 months from the date of sealing. Notes 1.

During the entire production process, operators at each station must wear an anti-static ring.

2. During operation, prevent the module from getting wet or dirty.

6.4 Recommended furnace temperature curve

Referred to IPC/JEDEC standard.

Peak Temperature : $<250^{\circ}\text{C}$

Number of Times : 2 times

